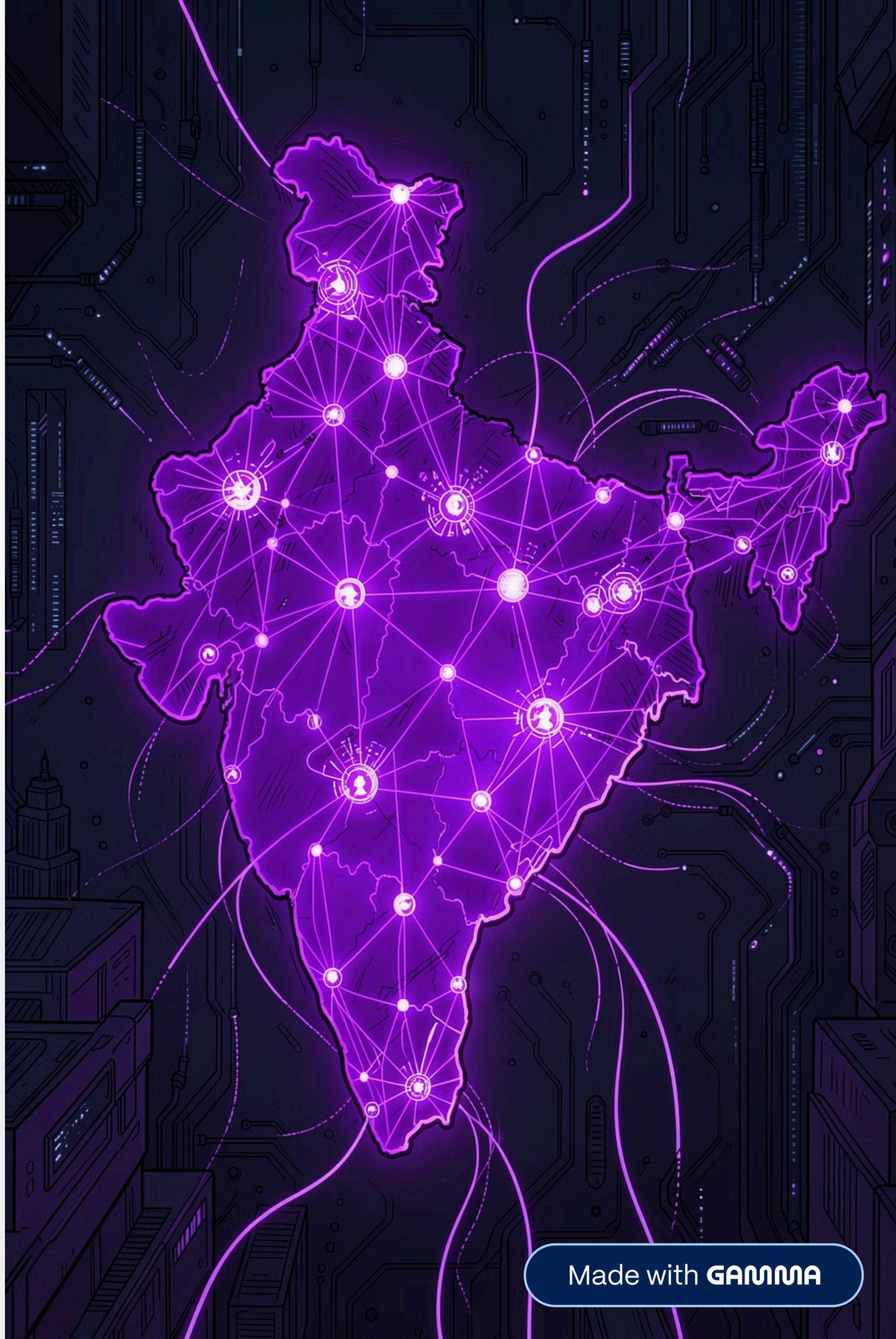
# AI-Powered Indian Cybercrime & Threat Intelligence Analytics Platform

An end-to-end intelligence platform using **Python, SQL Server, Power BI,** and AI integration to analyze 1,200 cybercrime incidents across India — transforming raw data into actionable threat intelligence.

CYBERSECURITY INDIA AI ANALYTICS



# The Growing Cyber Threat in India

With the rise of online banking, digital payments, e-commerce, and social media, India faces escalating cybercrime — causing significant financial losses and privacy risks.

### Common Attack Types

Ransomware, Phishing, Malware, Online Fraud, Identity Theft, Data Breaches, Hacking, Cyber Bullying

### Impact

Financial losses, privacy breaches, data theft, and operational risks across individuals, organizations, and public sectors

### Dataset

1,200 structured cybercrime incident records with year, city, attack type, sector, and financial loss in INR

# Data Dictionary

Each record captures a cybercrime case with the following fields:

| Column | Description |
| --- | --- |
| year | Year the case occurred |
| day | Day of the month |
| amount_lost_inr | Financial loss in Indian Rupees |
| incident_type | Type of cyber attack (ransomware, phishing, etc.) |
| city | Indian city where case was recorded |
| category | Affected sector |
| amount_lost_category | Low / Medium / High loss group |
| is_high_value_attack | Binary indicator for high-risk attack |
| day_range | Early, Mid, or Late month |
| case_date | Generated date field for analysis |

# Python EDA & Data Cleaning Pipeline

A 10-step cleaning pipeline prepared 1,200 records for analysis — from raw CSV to validated dataset.

01

### Import & Inspect

Load CSV, check shape (1,200 × 6) and data types

02

### Rename & Deduplicate

Standardize column names; zero duplicates found

03

### Clean Text

Strip spaces, title case; fix spellings (Banglore → Bangalore, Phising → Phishing)

04

### Fix Types & Missing Values

Convert types; fill numeric with median, categorical with "Unknown"

05

### Handle Outliers

IQR method to clip extreme financial loss values

06

### Feature Engineering

Derive amount_lost_category, is_high_value_attack, day_range, case_date

07

### Validate & Export

Final shape (1,200 × 10), zero nulls, exported to CSV

# SQL Server Threat Intelligence Analysis

13 SQL queries on **cyber_attacks** table (**cybercrime_analytics_db**) generated structured threat intelligence.

### Executive Summary

1,200 cases · ₹27.78 Cr total loss · ₹2,31,538 avg loss · 300 high-risk cases

### Yearly Trend (2019–2024)

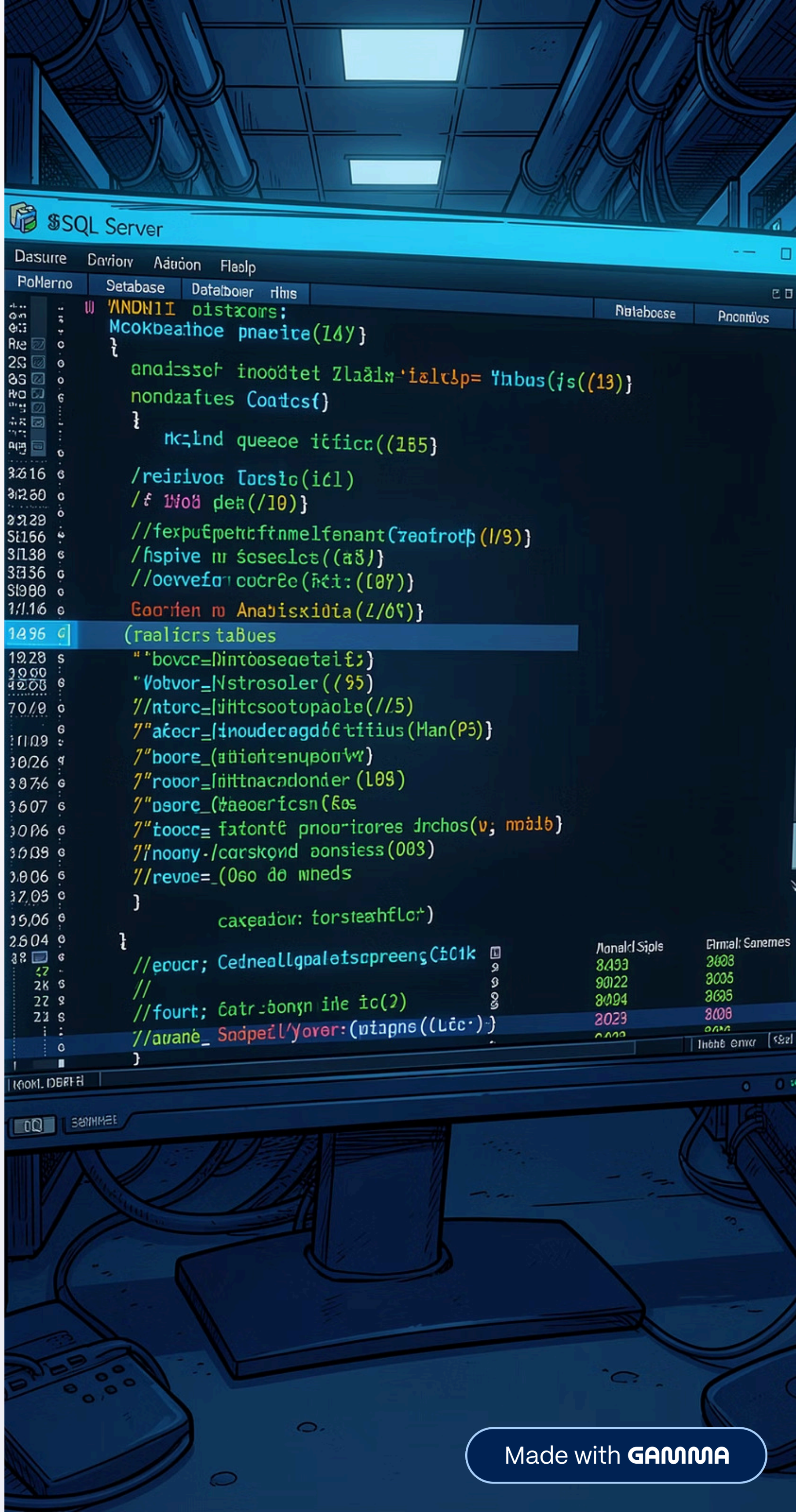
Peak loss in 2022 (₹5.09 Cr); 2024 recorded ₹4.51 Cr

### Top Attack Types

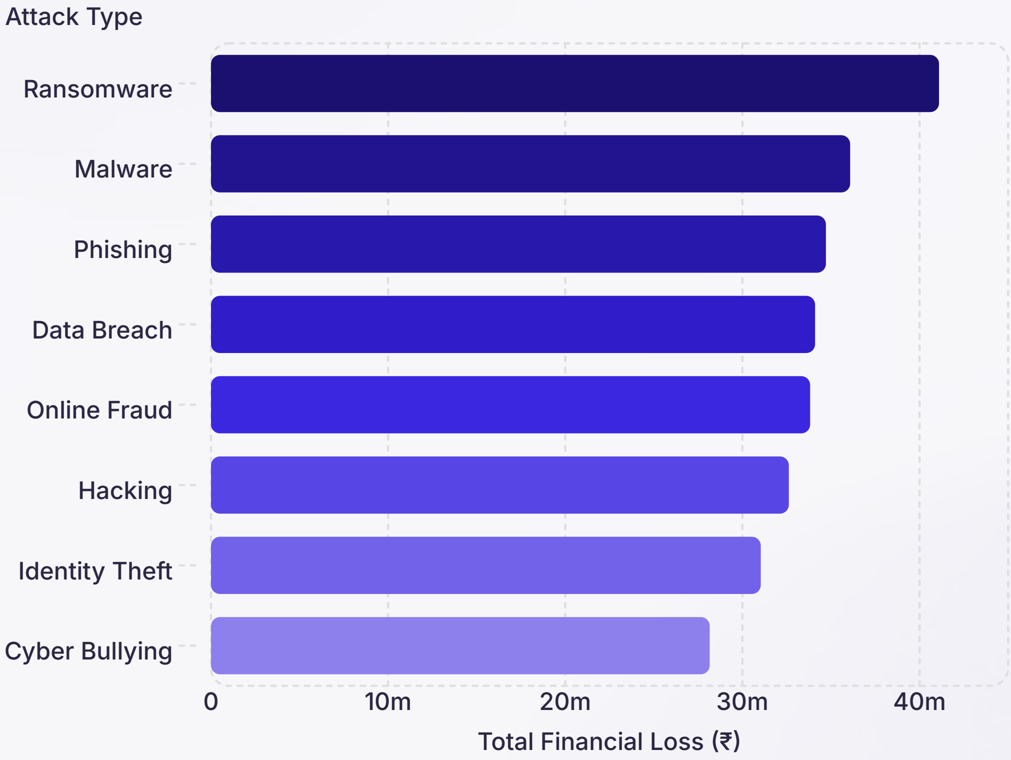
Ransomware led losses (₹4.10 Cr), followed by Malware (₹3.60 Cr) and Phishing (₹3.47 Cr)

### City & Sector Risk

Delhi ranked #1 (₹3.39 Cr); Social Media sector highest (₹3.89 Cr)



# Key SQL Insights: Attack Types & Financial Loss




## Attack Type Risk Analysis

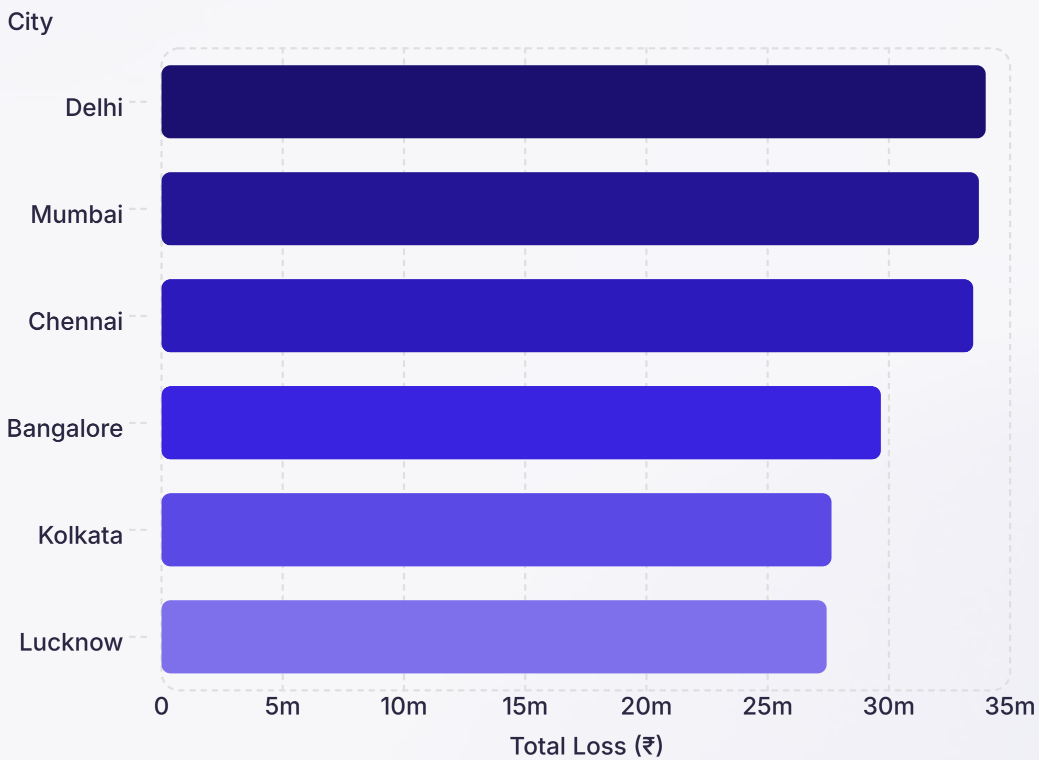
**Ransomware** caused the highest financial damage at ₹4.10 Cr across 189 cases. **Malware** and **Phishing** follow closely. All major attack types recorded between 116–189 incidents each, indicating a widespread and diverse threat landscape across India.

**Top 10 Highest Loss Incidents:** Individual losses ranged up to ₹4,99,946 — with Cyber Bullying, Ransomware, and Identity Theft appearing most frequently in the top-loss records.

# City-Level & Sector-Wise Risk Analysis

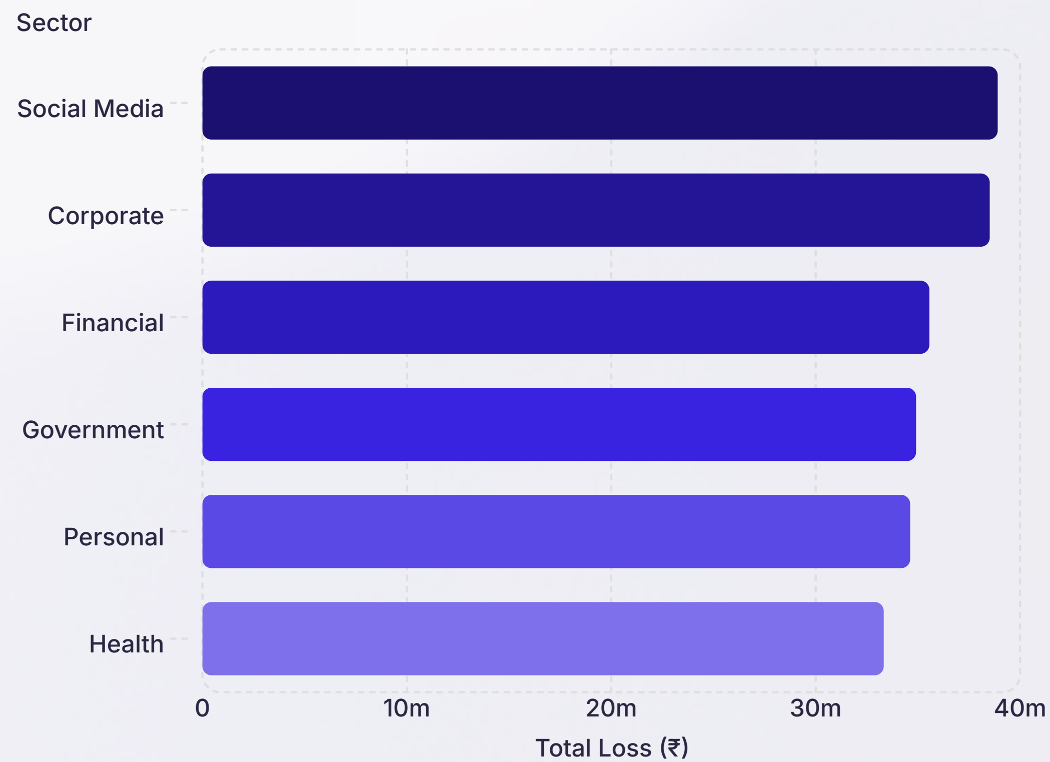
## Top Cities by Financial Loss

Delhi leads with ₹3.39 Cr across 143 cases. Mumbai, Chennai, and Bangalore follow — all major metros show consistently high cybercrime exposure with 30+ high-risk cases each.

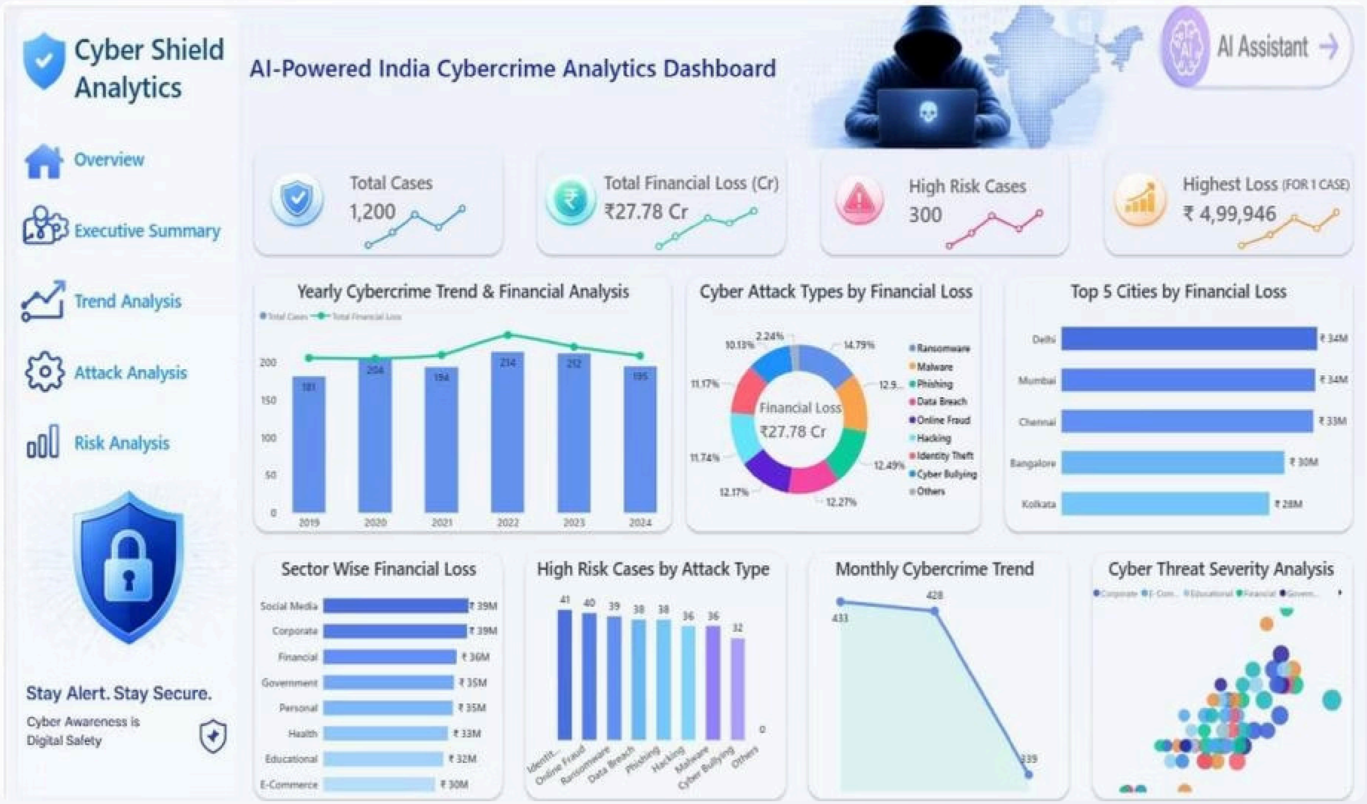



## Sector-Wise Financial Impact

**Social Media** (₹3.89 Cr) and **Corporate** (₹3.85 Cr) sectors face the highest losses. Financial, Government, and Personal sectors also show significant exposure — highlighting the need for sector-specific cybersecurity strategies.



# Power BI Dashboard: Cyber Shield Analytics



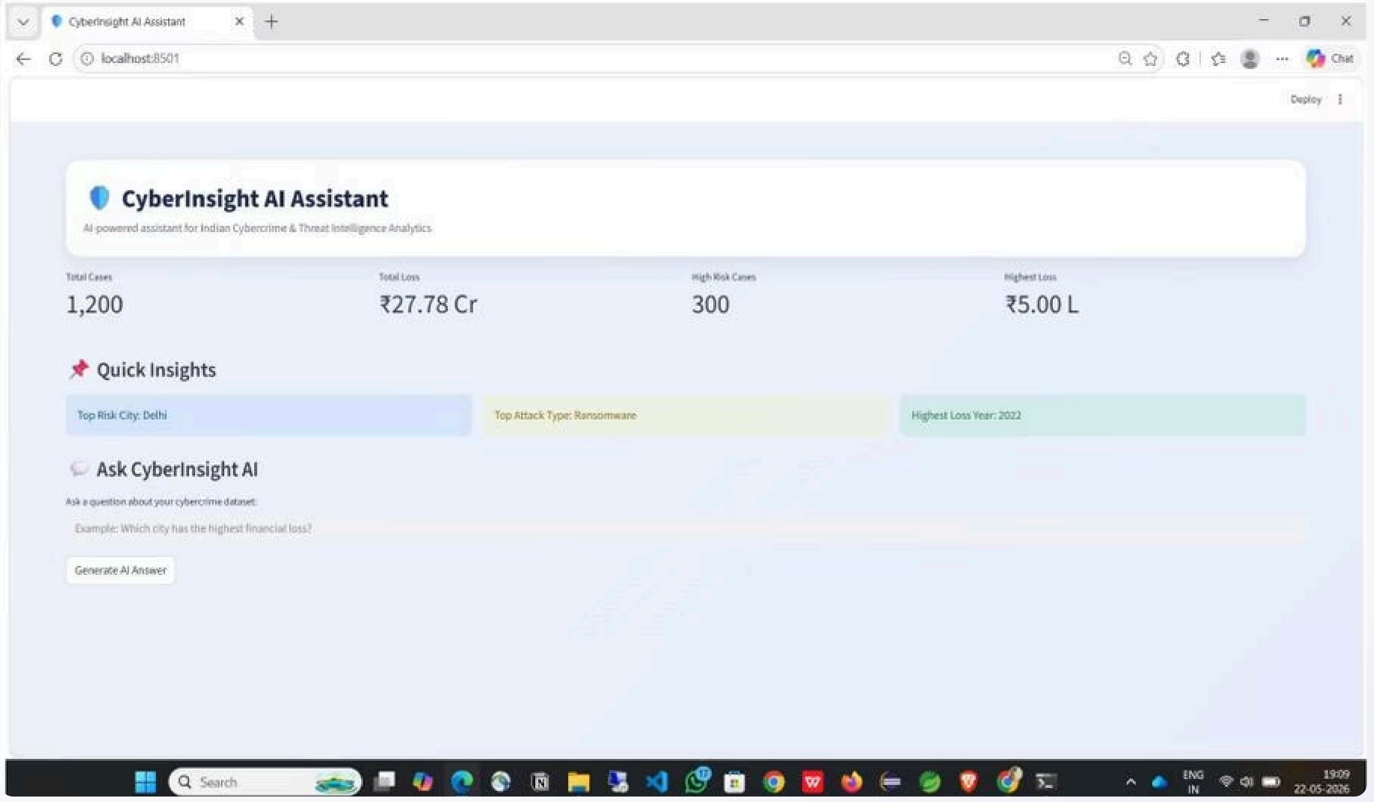

### KPI Cards

### Dashboard Visuals

- Yearly Cybercrime Trend & Financial Analysis
- Cyber Attack Types by Financial Loss (Donut)
- Top 5 Cities & Sector-Wise Loss (Bar Charts)
- High-Risk Cases by Attack Type
- Monthly Trend & Threat Severity Scatter Plot

| | | | |
|---|---|---|---|
| **1,200** | **₹27.78 Cr** | **300** | **₹4,99,946** |
| Total Cases | Total Financial Loss | High-Risk Cases | Highest Single Loss |

# CyberInsight AI Assistant & Project Architecture




## Streamlit AI Assistant

A **CyberInsight AI Assistant** provides an interactive query layer — displaying KPIs, quick insights (Top Risk City: Delhi, Top Attack: Ransomware), and answering natural language questions. Extensible via Gemini, Groq, or OpenRouter APIs.

## Project Architecture

### Raw Dataset

1,200 cybercrime records from CSV

### Python Cleaning

Preprocessing, feature engineering, export

### SQL Server

13 analytical queries for threat intelligence

### Power BI + AI

Interactive dashboard + Streamlit assistant

# Results, Conclusion & Future Scope

## Key Findings

1,200 cases · ₹27.78 Cr total loss · Ransomware highest-loss attack · Delhi, Mumbai, Chennai top-risk cities · Social Media & Corporate most affected sectors

## Conclusion

The platform successfully transforms raw cybercrime data into actionable intelligence using Python, SQL Server, Power BI, and AI — supporting cybersecurity awareness and data-driven decision-making.

## Future Scope

Real-time data feeds, ML-based attack prediction, anomaly detection, threat intelligence API integration, improved AI assistant with live model APIs, and online deployment.